# IPL Data Analysis Dashboard – Project Report

## 1. Project Overview

This project analyzes **Indian Premier League (IPL)** historical data to generate insights about team performance, player statistics, and match trends. The objective was to transform raw cricket datasets into meaningful insights using **Python for analysis** and **Power BI for visualization**.

statistics such as total runs, wickets, player performance, and match outcomes by selecting different seasons.

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## 2. Dataset Description

The analysis is based on two datasets:

**Matches Dataset**

- Contains match-level information such as season, teams, venue, toss decision, and match winner.
- Total records: **1095 matches**

**Deliveries Dataset**

- Contains ball-by-ball information including batsman, bowler, runs scored, and wickets.
- Total records: **260,920 deliveries**

These datasets were cleaned and transformed before performing analysis.

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## 3. Data Processing

The following preprocessing steps were performed:

- Standardized **season format** (e.g., 2007/08 converted to 2008).
- Corrected **team name inconsistencies** such as:
    - Kings XI Punjab → Punjab Kings

    - Delhi Daredevils → Delhi Capitals
- Handled missing values by filling city names, match methods, and dismissal fields.
- Created new metrics such as **total match runs, strike rate, and economy rate**.

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## 4. Dashboard Features

The Power BI dashboard presents multiple interactive visuals:

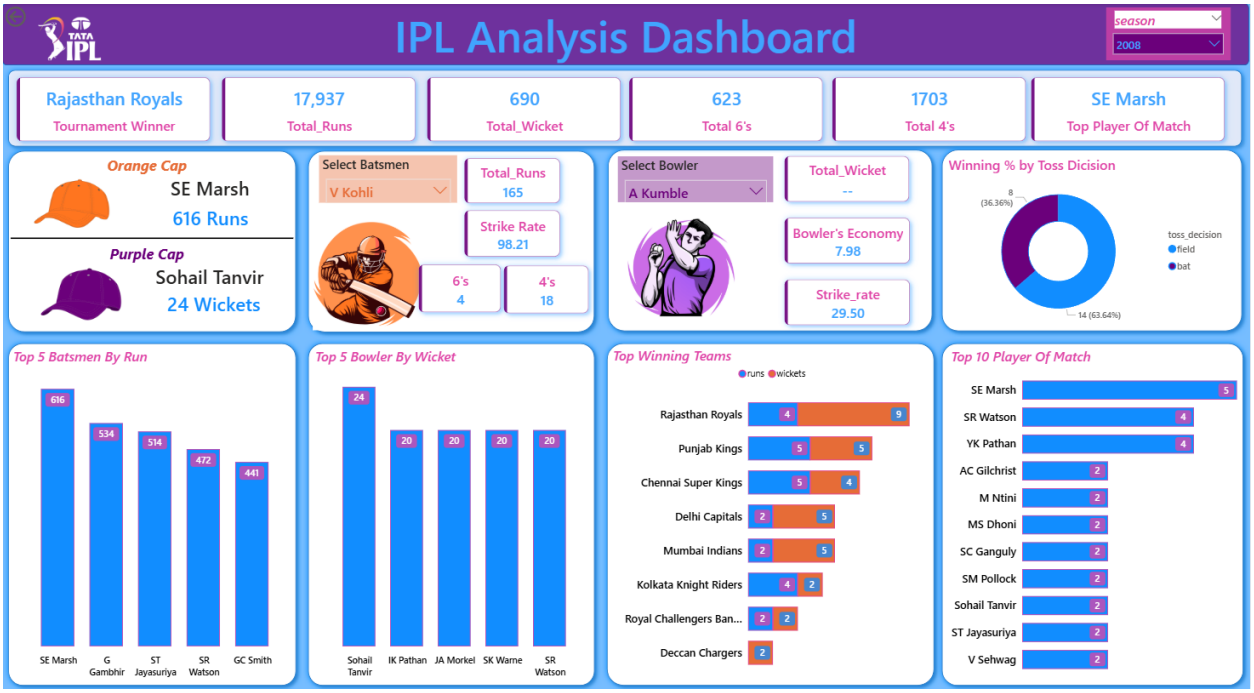
**Key Performance Indicators**

- Total Runs
- Total Wickets
- Total 4s and 6s
- Tournament Winner
- Top Player of the Match

**Player Performance Section**

- Select a batsman or bowler
- View runs, strike rate, boundaries, wickets, and economy rate

**Team Performance**



- Top winning teams
- Total runs scored by teams

**Top Players**

- Top batsmen by runs
- Top bowlers by wickets
- Top players with most “Player of the Match” awards

**Match Strategy Analysis**

- Win percentage based on **toss decision (bat vs field)**

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## 5. Key Insights

- Certain players consistently dominate run scoring across seasons.
- Some bowlers maintain very low **economy rates**, making them highly effective.
- Toss decisions influence match outcomes but do not guarantee victory.
- Certain stadiums produce **higher average match scores**, indicating batting-friendly conditions.

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## 6. Tools and Technologies

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Power BI**

Python was used for data cleaning and analysis, while Power BI was used to design the interactive dashboard.

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## 7. Conclusion

This project demonstrates how **sports analytics can convert large cricket datasets into meaningful insights**. By combining Python data processing with Power BI visualization, the dashboard provides an interactive platform to analyze IPL player performance, team success, and match patterns across seasons.

The project highlights practical skills in **data cleaning, exploratory analysis, and dashboard development**, which are essential for a **data analyst role**.